

AHB Master Feature Support Report

Summary of Supported and Unsupported Features
for the AHB Master



Prepared by:

Ahmed Yasser

1. Executive Summary

This report evaluates the implemented AHB master and its wrapper interface against the AHB-Lite transfer features visible in the submitted RTL. The purpose is to clearly identify which features are supported, partially supported, or not supported by the current implementation.

Overall conclusion: the design implements a simplified AHB master for 32-bit word transfers with IDLE, NONSEQ, and SEQ FSM states. The master also checks HREADY during transfer processing, allowing wait-state handling. However, several standard AHB features remain unsupported, including variable transfer sizes, BUSY transfers, complete fixed-length and wrapping bursts, error response handling, protection attributes, locked transfers, and multiple-slave decoding.

Category	Status	Summary
Basic read/write transfers	SUPPORTED	Basic single transfers can be initiated using the wrapper interface.
32-bit word transfers	SUPPORTED	HSIZE is fixed to 3'b010 and the data path is 32 bits wide.
Incrementing burst	PARTIALLY SUPPORTED	The RTL increments addresses by 4 only for HBURST = 3'b001.
Wait states	SUPPORTED	The master checks HREADY and can wait for transfer completion.
Protection attributes	UNSUPPORTED	HPROT is fixed to 4'b0000 and is not configurable..
Error response	UNSUPPORTED	HRESP is connected but the slave always drives it low and the master does not process errors.

2. Scope and RTL Architecture

The evaluated design consists of an AHB master, an AHB slave, a DATA_FIFO used for read data buffering, and an AHB wrapper. The wrapper connects the master directly to the slave and maps the external request interface to the AHB signals.

External request interface: W_ADDR_S, W_WRITE_S, W_BURST_S, W_ENABLE_S, and W_WDATA_S.

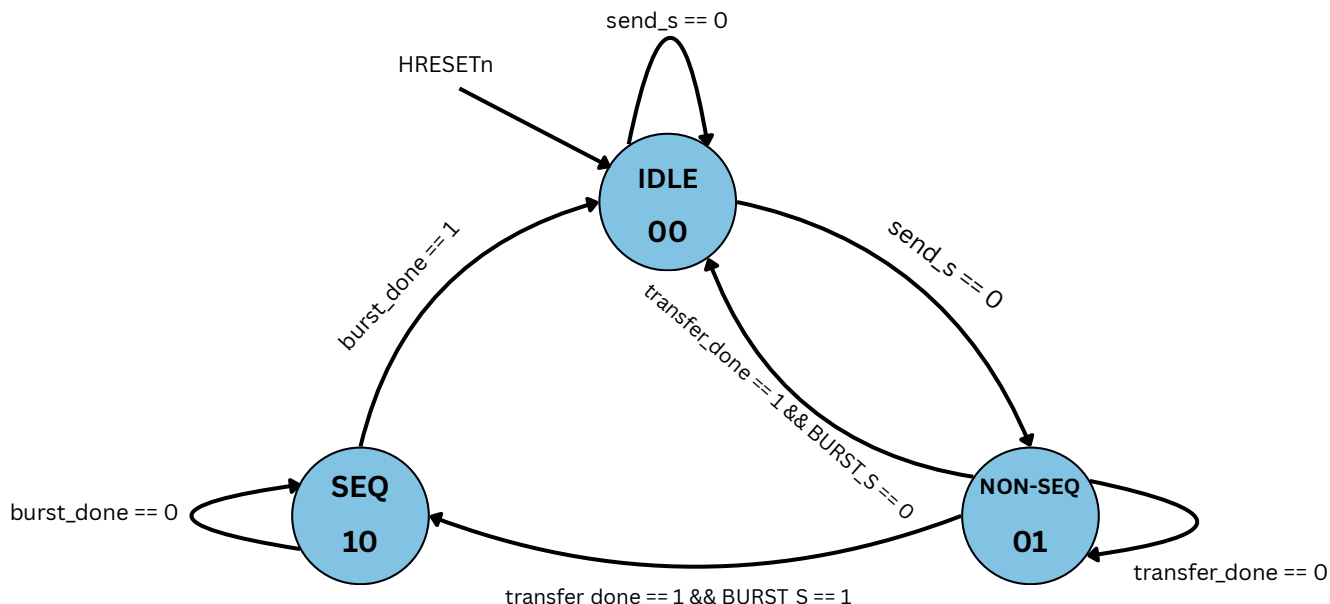
AHB signals generated by the master: HADDR, HWRITE, HSIZE, HBURST, HPROT, HTRANS, HMASTLOCK, and HWDATA.

Response signals received by the master: HRDATA, HREADY, and HRESP.

The wrapper uses W_ENABLE_S directly as HSELx. Therefore, the current system is a simple point-to-point master/slave connection rather than a multi-slave AHB interconnect.

3. AHB Master FSM

The master FSM contains three implemented states: IDLE, NON_SEQ, and SEQ. The next state is selected according to the external enable signal, transfer completion, and burst information.



IDLE: The master drives HTRANS = IDLE. When ENABLE_S is asserted, start_s becomes active and the FSM moves to NON_SEQ.

NON_SEQ: The first transfer of a transaction is issued with HTRANS = NONSEQ. The master drives the requested address and transfer controls. When the transfer completes, the FSM returns to IDLE for a single transfer or moves to SEQ for a burst.

SEQ: Subsequent transfers use HTRANS = SEQ. For the implemented incrementing burst case, the address is updated by four bytes for each 32-bit word transfer. When burst_done is asserted, the FSM returns to IDLE.

4. Feature Support Matrix

AHB Feature	Status	Evidence in RTL	Limitation
NONSEQ transfer	SUPPORTED	HTRANS = 2'b10 in NON_SEQ.	Used for the first transfer.
SEQ transfer	PARTIAL	HTRANS = 2'b11 in SEQ.	Address progression is explicitly implemented only for HBURST = 3'b001.
IDLE transfer	SUPPORTED	HTRANS = 2'b00 in IDLE.	No BUSY state is implemented.
BUSY transfer	UNSUPPORTED	No HTRANS = 2'b01 generation exists.	Cannot insert BUSY cycles.
32-bit data width	SUPPORTED	WIDTH = 32; HSIZE = 3'b010.	Transfer size is fixed to 32 bits.
Byte / halfword transfers	UNSUPPORTED	HSIZE is always 3'b010.	No variable-size transfer handling.
Single transfer	SUPPORTED	BURST_S = 0 causes return to IDLE after NONSEQ completion.	Basic single-transfer operation.
INCR burst	PARTIAL	ADDR_inc + 32'h0004 is used.	Only the HBURST = 3'b001 case is explicitly handled.
Fixed-length bursts	UNSUPPORTED	No burst-length counter or complete fixed burst handling.	INCR4/8/16 behavior is not implemented.
Wrapping bursts	UNSUPPORTED	No wrap-boundary calculation.	Address does not wrap at burst boundaries.

AHB Feature	Status	Evidence in RTL	Limitation
HREADY wait states	SUPPORTED	The master checks HREADY before completing transfer processing.	The integrated slave currently keeps HREADYOUT high when selected, so no wait is inserted by the provided slave.
HRESP error response	UNSUPPORTED	HRESP is tied low in the slave and not used by the master for error handling.	No ERROR response processing.
HPROT protection attributes	UNSUPPORTED	HPROT is always 4'b0000.	Protection attributes cannot be configured or used.
HMASTLOCK	UNSUPPORTED	HMASTLOCK is always 1'b0.	No locked transfer support.
Multiple slaves	UNSUPPORTED	W_ENABLE_S directly drives HSELx.	No address decoder or slave selection logic.

5. Detailed Supported Features

5.1 Basic Read and Write Transfers

The master accepts an address, read/write control, burst selection, enable signal, and write data. When enabled, the FSM moves from IDLE to NON_SEQ and drives the first transfer using HTRANS = NONSEQ.

For a write, HWDATA is driven toward the slave. For a read, the slave returns HRDATA, and the master stores the read data through the DATA_FIFO mechanism.

5.2 32-bit Word Transfers

HSIZE is permanently assigned 3'b010, corresponding to a 32-bit word transfer. The slave memory is also 32 bits wide.

5.3 Incrementing Address Operation

For the implemented sequential path, the next address is calculated as the current address plus four bytes. This is consistent with consecutive 32-bit word accesses because one word occupies four bytes. Example: 0x0000_0100, 0x0000_0104, 0x0000_0108, and 0x0000_010C.

6. Unsupported or Incomplete Features

6.1 Variable Transfer Sizes

Byte and halfword transfers are not supported. HSIZE is hard-coded to 3'b010, and the slave performs 32-bit memory accesses.

6.2 Complete Burst Support

The master does not implement the complete AHB burst encoding set. Sequential address update is explicitly guarded by BURST_S == 3'b001. Wrapping bursts and fixed-length bursts require additional boundary and length-control logic that is absent.

6.3 BUSY Transfers

The FSM generates IDLE, NONSEQ, and SEQ transfer types but never generates BUSY.

6.4 Response and Error Handling

HRESP is connected between the slave and master, but the slave always drives HRESP low and the master does not implement error-response processing. ERROR handling is therefore unsupported.

6.5 Protection and Locked Transfers

Protection attributes are not supported. HPROT is permanently driven to 4'b0000. HMASTLOCK is also permanently driven low, so locked transfers are not supported.

7. Wrapper-Level Limitations

The wrapper is designed for a simple point-to-point connection between one master and one slave.

W_ENABLE_S is directly connected to HSELx, so there is no address-based slave decoding. This is suitable for a simple test environment but is not a complete multi-slave AHB interconnect.

The wrapper exposes a simplified request interface. Several standard AHB signals are generated internally by the master and are not directly configurable by the external requester.

9. Final Conclusion

The submitted AHB master is a simplified 32-bit AHB-style master with a three-state FSM: IDLE, NON_SEQ, and SEQ. Its strongest supported functionality is basic word-based read/write operation, the NONSEQ-to-SEQ transfer flow, sequential word-address incrementing, and master-level HREADY wait-state handling.